

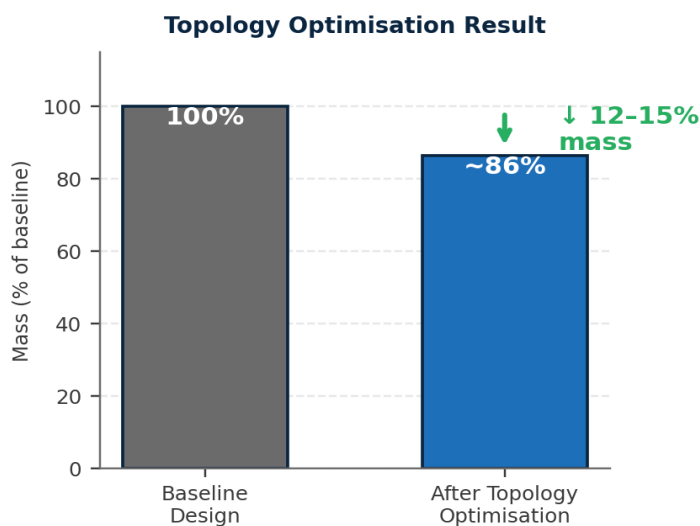
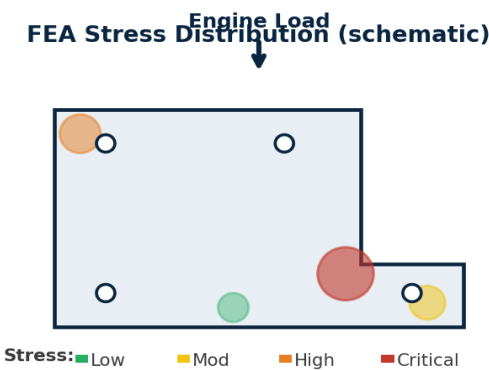
Aircraft Engine Bracket — Structural FEA and V&V

ANSYS-based simulation · 6+ load cases · Topology optimisation · ~12–15% mass reduction

SUMMARY

Developed and validated a structural simulation model of an aircraft engine bracket, evaluating response across **6+ load cases** including static, vibration, and thermal conditions. Applied FEA-based V&V comparing simulation outputs against reference behaviour and aerospace loading standards. Topology optimisation achieved **12–15% mass reduction** while maintaining full structural integrity within aerospace loading limits; identified critical stress concentration regions for design refinement and produced a formal technical report aligned with aerospace validation documentation standards.

KEY RESULTS	METHODOLOGY
Mass reduction ~12–15%	1. Geometry preparation — engine bracket CAD model imported from CATIA V5
Load cases evaluated 6+ (static · vibration · thermal)	2. Meshing strategy — refined elements at fastener holes & stress concentration zones
Structural integrity ✓ Maintained within aerospace limits	3. Load case definition — static aerospace loads + vibration spectra + thermal conditions
V&V approach Benchmarked vs reference behaviour & engineering standards	4. Linear and non-linear static FEA solved in ANSYS Mechanical
Critical zones Stress concentrations identified for design refinement	5. Topology optimisation — mass minimisation subject to stress & displacement constraints
	6. V&V — simulation outputs benchmarked against reference behaviour & aerospace standards
	7. Formal technical report — assumptions, methodology, V&V approach, design recommendations



TOOLS ANSYS Mechanical · ANSYS Topology Optimisation · CATIA V5 · Engineering V&V documentation

RELEVANCE Stress Analysis · Static & Damage Tolerance · Airframe Engineering · Engine Mounting · Aerospace V&V — applicable to roles at Airbus, Safran, GE Aerospace, Collins Aerospace, GKN, and aerospace ESPs